MANJOT SRAN

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RESEARCH INTERESTS

My research develops multimodal AI methods for reliable prediction across heterogeneous data modalities. I design and analyze fusion architectures - including early, mid, and late fusion; attention and gating mechanisms; mixture-of-experts models; and shared-private disentanglement - with emphasis on robustness to noise, missing modalities, and distribution shift. I also study representation learning under limited supervision and reliability-centered evaluation. In parallel, I integrate deep-learning-based fusion approaches with uncertainty-aware soft-computing techniques, particularly tolerance near sets, to quantify and manage uncertainty in multimodal classification and regression.

EDUCATION

University of Manitoba, Winnipeg, Canada *Starting Sept. 2026*
Incoming Ph.D. Student in Computer Science
Supervisors: Dr. Mengjun Hu and Dr. Sheela Ramanna

The University of Winnipeg, Winnipeg, Canada *Jan. 2024 – May 2026*
M.S. in Applied Computer Science
GPA: **4.375/4.5**
Thesis: *SPriG-TLC: A Novel Framework for Shared-Only Fusion with Private Feature Regulation and Tolerance-Based Ambiguity-Aware Clustering for Multimodal Affective Computing*
Thesis passed with distinction
Supervisor: Dr. Sheela Ramanna, Professor, Applied Computer Science Department, The University of Winnipeg
Relevant Coursework:

- Advanced Machine Learning
- Web and Document Databases
- Pattern Recognition
- Digital Image Processing

IKG Punjab Technical University, Jalandhar, India (IKGPTU) *June 2019 – June 2023*
B.Tech. in Computer Science and Engineering
Graduated with Distinction
CGPA: **8.83/10 (88.3%)**
Tier-1 accredited program by NBA (National Board of Accreditation, India)
Relevant Coursework:

- Data Structures
- Artificial Intelligence
- Digital Electronics
- Design and Analysis of Algorithms
- Machine Learning
- Database Management Systems
- Wireless Sensor Networks
- Computer Architecture and Microprocessor

AWARDS & HONORS

Graduate Studies Scholarship, The University of Winnipeg *May 2026*
Graduate Student Research Award (GSRA), The University of Winnipeg *April 2026*
- Competitive $7,000 graduate research award recognizing student-led research excellence.

Graduate Studies Scholarship, The University of Winnipeg *Aug. 2025*
Graduate Studies Scholarship, The University of Winnipeg *May 2025*
Graduate Studies Scholarship, The University of Winnipeg *July 2024*

SKILLS

Programming Languages:	Python, C, C++, JavaScript, Solidity
Web Development:	HTML, CSS, Bootstrap, Django, Flask, jQuery, XML
Networking:	Cisco Packet Tracer
Data Analysis:	Excel, Spreadsheets, SQL, NumPy, Pandas, Matplotlib
Machine Learning and Deep Learning:	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV, torchvision, librosa
LLM/GenAI Frameworks:	LangChain, LangGraph, LlamaIndex, OpenAI API, RAG, Vertex AI
Systems:	Linux, Git, Docker
Communication Skills:	Strong written and verbal communication skills

PUBLICATIONS

1. Manjot Singh Sran, Sheela Ramanna
SPriG: Shared-Only Fusion with Improvement-Guided Private Gating for Multimodal Affective Computing. *Information Fusion* (Elsevier), published online July 7, 2026, Article 104606. DOI: 10.1016/j.inffus.2026.104606 [Impact Factor: 17.4].
2. Manjot S. Sran, Sheela Ramanna, Ketan Kotecha
DiMoE: Disentangled Representation Learning with Mixture-of-Experts Fusion for Sentiment Intensity Prediction and Emotion Classification. *Algorithms* (MDPI), [under review], 2026.
3. Manjot S. Sran, Sheela Ramanna
TLC-STRIDE: An Ambiguity-Aware Framework for Multimodal Sentiment Prediction. Elsevier journal, [First Round of Revision Under Review], 2026.

RELEVANT EXPERIENCE

Teaching Assistant, Applications of Database Systems (ACS-2814-001) *Sept. 2025 – Dec. 2025*
University of Winnipeg - Winnipeg, Manitoba, Canada
- Led weekly labs and tutorials on ER modeling, normalization, and SQL (DDL/DML) with query design and optimization.
- Provided one-on-one support during office hours and troubleshot student issues with tools such as Access and SQL Server.
- Assisted with assessment administration (quizzes/exams) and delivered actionable feedback to improve learning outcomes.

Participant (Pathfinder Cohort) *June 2025 – Sept. 2025*
Winnipeg, Manitoba, Canada
- Participated in North Forge's Founders Program (Pathfinder cohort) to accelerate an early-stage digital health venture.
- Conducted user research with caregivers of people living with dementia to identify unmet needs, workflow gaps, and support preferences.

- Collaborated with mentors and peer founders to refine business model, value proposition, and go-to-market approach.
- Prepared prototype demos and delivered progress updates and pitches to North Forge stakeholders.

Senior Research Assistant *Aug. 2024 – Nov. 2024*

University of Winnipeg - Winnipeg, Manitoba, Canada

- Collaborated on the preparation and submission of an NSERC grant application, ensuring alignment with funding agency guidelines.
- Drafted and revised key proposal sections, including research objectives, methodologies, and projected impact.
- Managed deadlines and provided administrative and editorial support to finalize the application on time.
- Ensured high-quality content through iterative reviews and close communication with the lead researcher.

RELEVANT INDUSTRIAL TRAINING EXPERIENCE

Guru Nanak Dev Engineering College, Ludhiana, India *July 2021 – Aug. 2021*

Four Weeks of Industrial Training on Web Technologies

- Gained practical experience in web development technologies such as HTML, CSS, and JavaScript.
- Learned web development best practices, including responsive design, accessibility, and web security.
- Developed a website as part of the training program using HTML, CSS, JavaScript, and Bootstrap.

National Institute of Electronics & Information Technology (NIELIT), Chandigarh, India

July 2022 – Aug. 2022

Six Weeks of Industrial Training on Blockchain Technology

- Learned about blockchain technology and its practical applications.
- Developed smart contract creation skills using Solidity.
- Studied decentralized application development and the Ethereum platform.
- Worked in a team to develop a blockchain-based project as part of the training program.

TCIL-IT, Chandigarh, India *Jan. 2023 – June 2023*

Six Months of Industrial Training on Data Science and Machine Learning

- Gained practical skills in Data Science and Machine Learning.
- Explored and implemented machine learning libraries and algorithms.
- Learned how to fine-tune algorithm parameters and evaluate performance on real-world datasets.
- Completed a machine learning project on cricket score prediction as part of the training program.

RELEVANT PROJECTS

Project: Prompt-to-Video Generation Platform (SDXL + SVD with LLM Planning)

- Built a prompt-only AI video generation pipeline: user prompt → LLM-based scene planning → SDXL keyframe synthesis → Stable Video Diffusion rendering.
- Added reliability mechanisms including quality-gating and automatic fallback retries (seed and tuning adjustments) to improve coherence and reduce identity drift.
- Deployed an end-to-end service with FastAPI backend and Gradio UI for interactive video generation and result inspection.

Project: Feature Disentanglement + Mixture-of-Experts Fusion for Affective Computing

- Designed a shared-private disentanglement framework for multimodal affective computing.

- Implemented sparse top-k Mixture-of-Experts fusion with joint-router and separate-router variants.
- Evaluated on five benchmarks for sentiment intensity prediction and emotion classification across tri-modal and bi-modal settings.
- Performed ablation and interpretability analyses, including expert-usage and shared-versus-private contribution studies.

Project: Exploring Audio-Visual Fusion: Comparative Analysis of Fusion Strategies for Multi-Modal Emotion Recognition

- Explored audio-visual fusion using audio and video modalities.
- Conducted a comparative analysis of Early, Mid, and Late fusion strategies using Wav2Vec2 and VideoMAE.
- Achieved 95.39% accuracy on RAVDESS and 86.84% accuracy on CREMA-D with Late Fusion techniques.
- Demonstrated the strength of decision-level fusion approaches in multimodal tasks.

Project: KD-tree Construction and Search Algorithm Development

- Implemented KD-tree construction for managing k-dimensional data attributes.
- Developed Exact, Range, and Nearest Neighbor search algorithms.
- Optimized for speed and accuracy, comparing performance against linear search.
- Demonstrated significant efficiency gains with KD-trees over brute-force methods.

Project: Employee Management Systems Web Application

- Built with Python and Django framework.
- Implemented CRUD operations with Django ORM on SQLite.
- Designed a user-friendly UI for adding, deleting, filtering, and updating employee data.
- Ensured seamless data management and accessibility across the system.

RESEARCH SEMINARS & COLLOQUIA

Physics Colloquium, University of Winnipeg (organized by the Department of Physics)
Sept. 26, 2025
Winnipeg, Canada
Research Talk: Robust Alzheimer's Disease Detection via Multimodal Fusion

- Presented a multimodal AD diagnosis framework combining MRI (DeiT) with structured clinical variables (FT-Transformer), integrating modalities via early and mid fusion.

COMPETITIONS

North Forge RampUp 2025 — Pitch Competition *April, 2025*

- Pitched a startup idea in front of 300+ attendees, engaging judges and mentors.

3MT (Three-Minute Thesis) Competition *March, 2025*

- Presented "Advancing Emotion Recognition with Artificial Intelligence" at UWinnipeg's annual Three-Minute Thesis competition.

VOLUNTEERING

Volunteer, UWinnipeg STEM Days *May 7–8, 2025*

- Contributed to the STEM Days event organized by the University of Winnipeg.

- Delivered an outreach talk for Grade 7 students on May 7, 2025, presenting on Artificial Intelligence.

RELEVANT CERTIFICATIONS

- **Explainable Machine Learning (XAI)**, offered by Duke University *March 2025*
- **NLP: Twitter Sentiment Analysis**, certificate earned upon completion of a guided project offered by Coursera Project Network *March 2025*
- **Google AI Essentials**, offered by Google *Nov. 2024*
- **Deep Learning**, offered by Illinois Institute of Technology *Nov. 2024*
- **Critical Thinking for Better Decisions in the ChatGPT Era**, offered by Deep Teaching Solutions *Nov. 2024*
- **Data Science Math Skills**, offered by Duke University *Aug. 2024*
- **Deep Learning Applications for Computer Vision**, offered by University of Colorado Boulder *Aug. 2024*
- **Mathematics for Machine Learning: Multivariate Calculus**, offered by Imperial College London *Aug. 2024*
- **CS50 SQL Certificate**, offered by Harvard University *Dec. 2023*
- **CS50B Certificate**, offered by Harvard University *Sept. 2023*
- **CS50x Certificate**, offered by Harvard University *Aug. 2023*
- **Google Data Analytics**, offered by Google on Coursera *May 2023*
- **Python for Data Science and Machine Learning**, offered by Udemy *April 2023*
- **CS50's Intro. to Programming with Python**, offered by Harvard University *Feb. 2023*
- **Blockchain Technology Training Certificate**, offered by NIELIT, Chandigarh *Dec. 2022*

REFERENCES

Dr. Sheela Ramanna
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Department of Applied Computer Science
University of Winnipeg, Winnipeg, Canada
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Dr. Yangjun Chen
Full Professor
Department of Applied Computer Science
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